

Q1 Base case: When $n = 1$, then the only node is the root, so

$$p(r) = 1$$

$$\begin{aligned} \text{The formula, } 2^{(n+1)/2} - 1 \\ = 2^{(1+1)/2} - 1 = 1 \end{aligned}$$

Hence, $p(r) = 1$

Base case holds

IH: Assume that for all trees, T' , of size $n' \leq k$, for some $k \geq 1$, that for every node v' of T' , $p(v') \leq 2^{(n'+1)/2} - 1$

IS: Consider tree, T with $n = k+1$ nodes. To prove that for every node v in T , $p(v) = 2^{(n+1)/2} - 1$

Let y be the right most leaf in T , so, y has the largest possible level numbering in T . x is the ~~parent~~ parent of y .

Remove y and its siblings and form a ~~tree~~ tree T' with $n' = n - 2 = k - 1$ nodes.

$$\therefore p(v') \leq 2^{(n'+1)/2} - 1 = 2^{k/2} - 1 \quad [\text{from IH}]$$

$$p(y) = 2 \cdot p(x) + 1$$

$$p(x) \leq 2^{k/2} - 1$$

$$p(y) \leq 2(2^{k/2} - 1) + 1$$

$$p(y) \leq 2^{(k+2)/2} - 1 \rightarrow$$

since $p(v) \leq p(y)$ for all nodes v in T

$$p(v) \leq 2^{(n+1)/2} - 1$$

Conclusion: By strong induction,

$p(v) \leq 2^{(n+1)/2} - 1$ holds true for all nodes in a binary tree T with n nodes.

Q2 Algorithm has Same Elements (A, B)

```
if length(A) != length(B) then  
    return False  
end
```

```
counts ← array of size  $2n$ , initialised to 0  
for  $i \leftarrow 0$  to  $\text{length}(A) - 1$  do  
    if  $A[i] \geq 1$  and  $A[i] \leq 2n$  then  
        counts[A[i]] += 1  
    end  
end
```

```
for  $i \leftarrow 0$  to  $\text{length}(B) - 1$  do  
    if  $B[i] \geq 1$  and  $B[i] \leq 2n$  then  
        counts[B[i]] += 1  
    end  
end
```

```
for i ← 1 to 2n do
    if counts[i] ≠ 0
        return False
    end
end

return True
```

Runtime Justification: The total runtime for

this code is $O(n)$. The if statements
are $O(1)$ and the for loops running
are $O(n)$.

CSC Assignment 4

Q3 $S = [5, 3, 17, 10, 84, 19, 6, 22, 9]$

Step 1: Build a max heap

1 Start at $S[3] = 10$ swap with 22

2 Heapify $S[2] = 17$ swap with 19

3 Heapify $S[1] = 3$ swap with 84

4 Heapify $S[0] = 5$ swap with 84

Max heap:

$S = [84, 22, 19, 5, 3, 17, 6, 10, 9]$

Step 2: In-place sorting

1 Swap 84 with 9

Reduce heap size to 8 and heapify $S[0]$

compare 9 to 22 and 19.

swap 22 and 9.

heapify $S[1] = 9$, swap 9 with 10

2 Swap 22 with 9

$S = [9, 10, 19, 5, 3, 17, 6, 22, 84]$

Reduce Heap size to 7 and
heapify $S[0]$

compare 9 to 10 and 19.
Swap 9 with 19.

$S = [19, 10, 9, 5, 3, 17, 6, 22, 84]$

Heapify $S[2]$: No swaps needed

3 Swap 19 with 6

$S = [6, 10, 9, 5, 3, 17, 19, 22, 84]$

Reduce heap size to 6 and heapify
 $S[0]$

compare 17 to 6 and 9.
Swap 17 and ~~19~~ 6

$S = [17, 10, 9, 5, 3, 6, 19, 22, 84]$

Repeat for remaining elements

Sorted array = $[3, 5, 6, 9, 10, 17, 19, 22, 84]$

4 swap 17 with 6

Reduce heap size to 5

Heapify $s[0] = 6$

compare 6 to 10 and 9

swap 6 with 10

Heapify $s[1] = 6$

no swaps needed

$S = [10, 6, 9, 5, 3, 17, 19, 22, 24]$

5 swap 10 with 3

Reduce heap size to 4

Heapify $s[0] = 3$

compare 3 to 6 and 9

swap 3 with 9

Heapify $s[2] = 3$, no swaps needed

$S = [9, 6, 3, 5, 10, 17, 19, 22, 24]$

6 swap 9 with 5

$S = [5, 6, 3, 9, 10, 17, 19, 22, 24]$

Reduce Heap size to 3

Heapify $S[0] = S$

compare 5 to 6 and 3

swap 5 with 6

$S = [6, 5, 3, 9, 10, 17, 19, 22, 24]$

7 swap 6 with 3

$S = [3, 5, 6, 9, 10, 17, 19, 22, 24]$

Reduce heap size to 2

Heapify 3

compare 3 to 5

swap 3 and 5

$S = [5, 3, 6, 9, 10, 17, 19, 22, 24]$

8 swap 5 with 3

reduce heap size to 1

no need to heapify

Final sorted array is ready.

$S = [3, 5, 6, 9, 10, 17, 19, 22, 89]$

Q4 Prove that $\ln(n!) \in \Theta(n \ln(n))$

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n \frac{1}{e^{\frac{1}{12n+1}}} < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$

Taking log on each side and then splitting it to upper and lower bounds

upper: $\ln(n!) < \ln(\sqrt{2\pi n}) + \ln\left(\frac{n}{e}\right)^n + \ln(e^{\frac{1}{12n}})$

lower: $\ln(n!) > \ln(\sqrt{2\pi n}) + \ln\left(\frac{n}{e}\right)^n + \ln(e^{\frac{1}{12n+1}})$

Simplify each term:

upper: $\ln(n!) > \frac{1}{2} \ln(n) + n(\ln(n)) - n$

lower: $\ln(n!) < \frac{1}{2} \ln(n) + n(\ln(n)) - n$
+ negligible terms

The dominant terms in both bounds is $n(\ln(n))$

By proving the bounds, we have proven that $\ln(n!) \in \Theta(n \ln(n))$

Q5 $x_i \in [1, 2, 3]$

